

# PUF + SNN Security for Quest 3 Motion Data

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## Problem & Research Question

## **The Problem**

Quest 3 applications depend on a continuous stream of motion data.

An attacker may

- Replay data from an earlier session
- Modify motion data or its metadata
- Submit data under the wrong device or session
- Produce abnormal motion before it is authenticated

**Can a software prototype using a simulated noisy PUF-derived session credential reject replayed, modified, or misattributed Quest 3 head-motion windows before SNN inference while meeting predefined attack-rejection, macro-F1-loss, and latency targets?**

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Pilot Contribution & Scope

## What this pilot studies

- Quest 3 head position and quaternion orientation
- Five controlled head-motion classes
- Simulated PUF authentication before inference
- Conventional classifiers compared with an SNN
- Separate detection of abnormal sensor patterns
- Attack rejection, model performance, and processing latency

## Initial contribution

A reproducible software testbed that shows which attacks are handled by authentication, which require motion analysis, and what each layer costs.

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Literature & Technical Landscape:  
XR & SNN

| Area                  | Findings                                                                     | Relation to our Project                                        | Source                                     |
|-----------------------|------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------|
| Motion privacy        | Head and hand motion can identify VR users                                   | Treat motion as sensitive data and prevent split leakage       | Nair et al. [1]                            |
| Tracking attacks      | XR motion streams can be blocked or manipulated                              | Test freezes, missing samples, noise, and pose jumps           | Rafique and Cheung [6] & Shoaib et al. [7] |
| SNN classification    | SNNs can learn temporal motion & activity patterns                           | Test a small recurrent LIF classifier                          | Eshraghian et al. [5] & Li et al. [9]      |
| SNN anomaly detection | SNNs can detect unusual time-series patterns but aren't automatically robust | Evaluate a separate detector and compare clean vs altered data | Bäßler et al. [8] & Sharmin et al. [12]    |

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Literature & Technical Landscape:  
PUF & Authentication

| Area                      | Findings                                                                                       | Relation to our Project                                                                            | Source                 |
|---------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------|
| XR authentication         | PUF-derived keys can support lightweight XR authentication and session-key establishment       | Use a simulated PUF-derived credential within the software prototype                               | Cha et al. [2]         |
| PUF modeling attacks      | Exposed challenge-response behavior can be modeled by ML                                       | Modeling attacks will be optional until PUF interface has been fully defined                       | Rührmair et al. [3]    |
| Noisy PUF response        | PUF outputs must have stability before key reconstruction can be reliable                      | Measure BER, reliability, uniqueness, and uniformity first, then error correction can be justified | Herder et al. [4]      |
| Authentication Boundaries | Authentication mechanisms protect certain trust properties, not the meaning of the sensor data | Keep the authentication gate separate from abnormal-motion detection                               | Stephenson et al. [10] |

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Research Gap

## Targeted-review finding

Based on our review to date, we have not identified a study that evaluates all of the following in one reproducible framework:

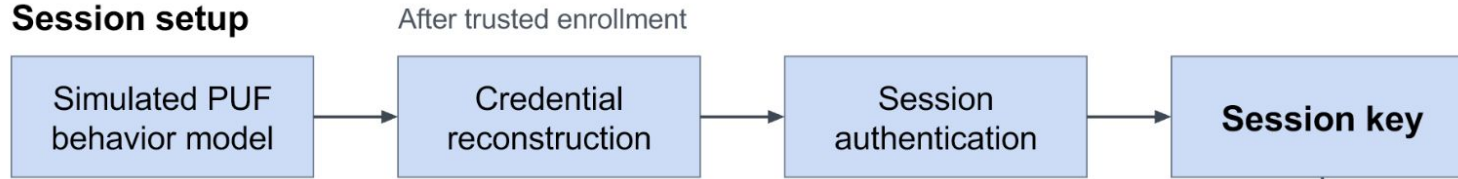
- Simulated PUF-derived device and session authentication
- Authenticated Quest head-motion windows
- SNN motion classification and separate abnormality detection
- Replay, cross-session, cross-device, and sensor-stream attacks
- Per-stage accuracy, security, and latency measurements

**Our focus is the interaction between these parts, not simply the individual technologies.**

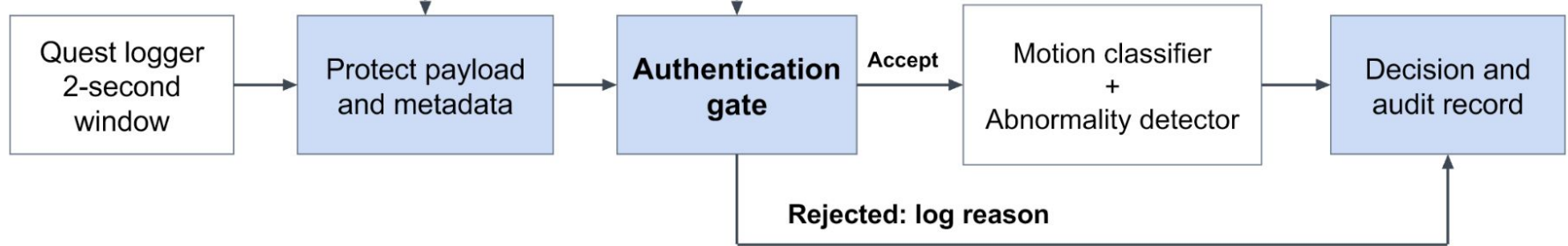
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System Architecture & Trust Flow

## Session setup



## Window processing



Only accepted windows reach the models. Accepted data may still contain abnormal motion.

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Threat Model

### Trusted for the Pilot

- Enrollment data and verifier
- Implementation of authentication
- Quest logger and window builder
- Local experiment/config pipeline

### Out of Scope/Limitations

- No physical PUF in Quest 3
- No compromised verifier
- Rooted or modified Quest firmware
- Advanced adaptive PUF attacks are deferred
- No firmware or side-channel attacks

### Attacker Capabilities

- Replay a previous valid window
- Substitute a window from a separate device or session
- Change protected metadata or payload
- Replay, delay, drop or reorder windows
- Inject abnormal sensor behavior

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Head Motion Tasks & Trials

## **Initial sensing task**

Quest 3 head-motion classification

## **One trial**

1 second prompt → 2 second action → 1 second rest

## **Initial sampling target**

60 Hz × 2 seconds = 120 samples per window

- **Nod**
- **Shake**
- **Look left and return**
- **Look right and return**
- **Still**

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## Sensor Window & Authentication Context

## Sensor and Experiment Data

- Head position (x, y, z)
- Orientation Quaternion (x, y, z, w)
- Capture Timestamp
- 120 time steps over fixed 2-second window
- Quest tracking-valid flag
- Trial-relative position/orientation
- Participant, session, and trial IDs

## Authentication Context

- Device ID
- Session ID
- Window ID and increasing sequence number
- Start and end time of window
- Data-format/version identifier
- Entire processed motion payload
- Protocol and schema versions
- Authentication tag and gate result

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Processing, Dataset, & Splits

## **Processing**

- Device origin coordinates
- Position and orientation relative to the first valid trial pose
- Quaternion normalization and sign continuity
- Resample to 120 time steps
- Reject windows with large gaps or poor tracking

## **Primary split**

Session 1: training

Session 2: validation

Session 3: testing

## **Initial synthetic dataset**

6 device profiles  $\times$  3 sessions  $\times$  20 trials  $\times$  5 classes = 1,800 clean windows

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Simulated PUF & Authentication  
Design

## Simulation PUF Pipeline

1. PUF behavior model
  - Simulate device specific ring oscillator PUF responses
  - Include manufacturing variation and repeated-read noise
  - Create an enrollment reference response
  - Measure BER, reliability, uniqueness, and uniformity
2. Credential reconstruction
  - Use the measured PUF behavior to reconstruct a stable secret
  - Select error-correction methods based on measured BER
  - Measure reconstruction success and false rejection
3. Session and Window Authentication
  - Derive an authentication key from the reconstructed secret
  - Protect each motion window using HMAC-SHA-256
  - Bind the payload to device, session, sequence, timing, and format
  - Reject invalid, stale, duplicate, or reordered windows before inference

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## Classification & Abnormal Behavior Detection

| Component              | Initial Method                      | Purpose                                    |
|------------------------|-------------------------------------|--------------------------------------------|
| Conventional baselines | Logistic regression & random forest | Establish simple, reproducible performance |
| SNN classifier         | Recurrent LIF network               | Predict the motion class                   |
| Abnormality baseline   | Supervised binary model             | Predict normal or suspicious behavior      |
| Later additions        | LightGBM & SNN abnormality model    | Added after initial baselines work         |

The classifier and abnormal behavior detector are evaluated separately

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## Attack Plan

| <b>Tier</b>                      | <b>Attack Type</b>                 | <b>Examples</b>                                                     | <b>Expected Defender</b> |
|----------------------------------|------------------------------------|---------------------------------------------------------------------|--------------------------|
| Tier 1 – Authentication          | Replay / Wrong Identity            | Old window, wrong session, wrong device, duplicate sequence.        | Authentication gate      |
| Tier 1 – Authentication          | Protected-window modification      | Change pose data, IDs, timestamps, sequence number or metadata.     | HMAC verification        |
| Tier 1 – Authentication          | Session-authentication failure     | Invalid and stale session credential or a failed PUF reconstruction | Session Authentication   |
| Tier 2 – Semantic Sensor Attacks | Abnormal but validated motion data | Noise, drift, jitter, missing samples, freezes                      | Abnormality detector     |
| Tier 3 – Later Attacks           | Adaptive PUF attacks               | PUF modeling or attacks to mislead the classifier                   | Future evaluation        |

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Controlled System Comparison

- 1. Conventional classifier without authentication**
- 2. SNN classifier without authentication**
- 3. Simulated-PUF-authenticated stream  
with a conventional classifier**
- 4. Simulated-PUF-authenticated stream  
with an SNN classifier**

### **Controlled comparison**

Same data, splits, attacks, preprocessing, and random seeds.

Only the authentication and inference conditions change.

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What Defines Success?

The 2 second collection window is not included in the 20 ms target for latency

### **Security**

- 100% observed rejection across at least 1,000 cases per Tier-1 attack
- Valid-window authentication false-reject rate  $\leq 0.5\%$

### **Classification**

- Authentication-related macro-F1 loss  $\leq 1$  percentage point
- SNN macro-F1 within 5 percentage points of the best baseline

### **Abnormality detection**

- $\geq 90\%$  detection for medium/severe conditions at  $\leq 5\%$  clean FPR

### **Latency**

- Authentication p95  $\leq 1$  ms per window
- Post-window decision p95  $\leq 20$  ms
- Authentication overhead  $\leq 10\%$

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Tools & Development Environment

| Area                 | Initial Tools                                    |
|----------------------|--------------------------------------------------|
| Quest development    | Quest 3, Unity 6 LTS, OpenXR 1.16.1              |
| Deployment           | Android Build Support, ADB, Meta developer setup |
| Data processing      | Python, NumPy, pandas, SciPy                     |
| ML baselines         | Scikit-learn logistic regression & random forest |
| Later ML baseline    | LightGBM                                         |
| SNN development      | PyTorch & snnTorch                               |
| Analysis             | Matplotlib, Seaborn, scikit-learn metrics        |
| Collaboration        | Git, GitHub, VS Code                             |
| PUF & authentication | Python, NumPy, HMAC-SHA-256                      |

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Proposed Responsibilities

## Keegan: XR, data, and inference

- Quest logger and motion task
- Schema validation and preprocessing
- Synthetic data and Tier-2 transforms
- Train/test splits and leakage checks
- Logistic regression and random forest
- SNN classifier and anomaly baseline

## Will: PUF and authentication

- Configurable simulated PUF
- PUF reliability and BER tests
- Enrollment and reconstruction
- Session and window authentication
- Tier-1 authentication attacks
- Authentication metrics

## Shared

Research question and scope, threat model, authenticated-window interface, four-condition experiment, integration and audit records, literature, presentation, and analysis.

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1 Month Deliverables

**Week 1.** A validated Quest-style logger or generator that produces the agreed schema and passes automated checks

**Week 2.** A measured simulated PUF model with reliability, uniqueness, uniformity, BER, and reconstruction results

**Week 3.** Reproducible logistic regression and random forest baselines, followed by an initial SNN classifier and abnormality baseline

**Week 4.** An end-to-end simulated-PUF-gated inference run with Tier-1 attacks, audit records, and per-stage latency

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Out of Scope for the Pilot

### **What Is Explicitly Out of Scope for the Pilot**

- Claiming that the Quest 3 contains or exposes a usable PUF
- FPGA or Loihi deployment during the first month
- Eye tracking or hand-joint tracking
- Rooted Quest firmware or raw sensor firmware access
- A production authentication server
- Full payload confidentiality unless later justified
- Compromised-verifier and hardware side-channel attacks
- Advanced PUF-modeling or adaptive adversarial attacks

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Week 2 Progress: Keegan

## **Synthetic head motion pipeline**

- Generated 1,800 Quest-like motion windows
- Five balanced motion classes
- Six simulated devices and three sessions per device
- Separate training, validation, and testing sessions
- Added schema, data-quality, and leakage checks

## **Results**

- All 1,800 windows passed validation
- All five automated tests passed
- Generator produced the same results when rerun with the same seed

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Week 2 Progress: Will

## Completed Tasks

- Built a configurable simulation of RO-PUF in Python
- Modeled persistent device variations and measurement noise
- Used 128 ring oscillators with adjacent pairing to generate 64-bit response
- Added an enrollment reference and repeated noisy PUF response
- Measured raw BER, reliability, uniqueness, and uniformity
- The simulation outputs plots, for same vs different device behavior, BER vs noise, and response uniformity.
- Random seeds are reproducible two identical seeds yield the same results
- Validated the designs
- 6 Simulated devices
- 128 ROs per device
- Adjacent RO pairing
- Config-driven experiment
- Reproducible runs

## Next Step

Use the PUF behavior to design and test stable reconstruction before session/window authentication

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Faculty Decisions Needed

## Current proposed approach

- Continue documentation, synthetic-data testing, and software simulation on personal laptops
- Don't deploy to lab headsets or retain human-derived data until the required permissions are confirmed
- Use cross-session performance as the primary synthetic-pilot evaluation
- Report median and p95 latency with provisional targets of:
  - Authentication  $p95 \leq 1$  ms
  - Post-window decision  $p95 \leq 20$  ms
  - Authentication overhead  $\leq 10\%$
- Delay FPGA work until the software pipeline produces results

## Decisions needed now

- Confirm which synthetic, public, or scripted-device activities are permitted before the institutional human-subjects determination
- Review whether our provisional latency targets fit the intended XR application

## Questions we will bring later

- Headset access and computer workflow—after completing the inventory and setup checklist
- Data storage and retention—after drafting the data-management plan
- FPGA access—after completing the software milestone and researching hardware options

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References

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